

Estimated Annual Agricultural Pesticide Use Jefferson County, Idaho 2014-2018

24 July 2026

Summary

Jefferson County ranks **#4 of Idaho's 44 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of ID's five highest pesticide-use counties. Jefferson County, in eastern Idaho near Rigby, grows irrigated potatoes, wheat, and barley alongside dairy operations.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Jefferson County, Idaho** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound
OTH	Other

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Jefferson County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM POTASSIUM	DTC	739900	339268	236124
2	METAM	DTC	345594	512798	531684
3	SULFURIC ACID	INO	143804	270539	295180
4	CHLOROPICRIN	OC	142419	63096	47333
5	DICHLOROPROPENE	OC	76178	120436	106961
6	GLYPHOSATE	OTH	12681	11942	13496
7	PENDIMETHALIN	OTH	10572	10943	9312
8	CHLOROTHALONIL	OTH	10462	9687	8574
9	METRIBUZIN	TRZ	10098	9763	8845
10	2,4-D	CPX	9177	6378	5635
11	EPTC	TC	6401	8101	8322
12	MCPA	CPX	6039	4425	5940
13	MALEIC HYDRAZIDE	OTH	4978	3264	3739
14	DIMETHENAMID & DIMETHENAMID-P	OTH	4183	4934	4009
15	DIMETHENAMID-P	OTH	4183	4401	3690
16	PHOSPHORIC ACID	INO	3914	3834	6128
17	METOLACHLOR & METOLACHLOR-S	OTH	3806	3153	2901
18	METHOMYL	CB	3664	3080	2381
19	FLUROXYPYR	OTH	3627	3061	2916
20	CHLORPYRIFOS	OP	3501	2376	2133
21	METOLACHLOR-S	OTH	3460	2772	2424
22	MANCOZEB	DTC	3445	3673	6728
23	BROMOXYNIL	OTH	2990	2475	3275
24	ACETOCHLOR	OTH	2127	901	1008
25	QUINTOZENE	OTH	1973	1097	1097
26	PYRIMETHANIL	OTH	1838	1570	1506
27	DIQUAT	DIP	1638	1576	1892
28	ATRAZINE	TRZ	1547	1977	1226
29	PROPARGITE	OTH	1389	1389	2135
30	GLUFOSINATE	OTH	1339	683	531
31	BOSCALID	OTH	1268	1324	1370
32	PINOXADEN	OTH	1242	1111	1046
33	HEXAZINONE	TRZ	1222	1284	850
34	AZOXYSTROBIN	OTH	1154	993	1045
35	IMIDACLOPRID	OTH	1090	1080	1228
36	DIURON	OTH	1022	1239	1077
37	PARQUAT	DIP	995	1718	4376
38	DICAMBA	OTH	917	601	483
39	FLUOPYRAM	OTH	874	728	599
40	DIMETHOATE	OP	839	2510	2688
41	PYRACLOSTROBIN	OTH	794	634	658
42	OXAMYL	CB	716	686	2388
43	TRIFLURALIN	OTH	550	828	484
44	FLUTOLANIL	OTH	444	1129	1108
45	PROPICONAZOLE	OTH	403	351	379
46	CLOPYRALID	OTH	374	296	386
47	METOLACHLOR	OTH	346	380	478

Table 2: Top 80 EPest-high Estimates, Jefferson County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
48	MEFENOXAM	OTH	333	482	445
49	CYFLUTHRIN	PYR	320	158	105
50	CARFENTRAZONE-ETHYL	OTH	319	224	151
51	CLETHODIM	OTH	310	157	214
52	METIRAM	DTC	309	309	547
53	FLUXAPYROXAD	OTH	295	203	153
54	PICLORAM	OTH	290	103	63
55	CYHALOTHRIN-LAMBDA	PYR	255	297	235
56	PROTHIOCONAZOLE	OTH	250	143	110
57	TEBUCONAZOLE	OTH	250	139	88
58	TRI-ALLATE	TC	247	206	261
59	FLUAZINAM	OTH	241	244	397
60	DIFENOCONAZOLE	OTH	226	178	263
61	2,4-DB	CPX	221	338	295
62	METALAXYL	OTH	202	202	442
63	THIFENSULFURON	OTH	199	209	161
64	MANDIPROPAMID	OTH	186	148	189
65	HEXYTHIAZOX	OTH	163	86	103
66	SPIROTETRAMAT	OTH	162	136	84
67	ALPHA CYPERMETHRIN	PYR	161	107	107
68	CYPROCONAZOLE	OTH	157	93	93
69	COPPER HYDROXIDE	INO	120	401	423
70	TRIBENURON METHYL	OTH	109	104	90
71	LINURON	OTH	107	114	114
72	ZETA-CYPERMETHRIN	PYR	104	95	87
73	PICOXYSTROBIN	OTH	90	65	94
74	PYRASULFOTOLE	OTH	88	65	60
75	BIFENTHRIN	PYR	85	107	120
76	PETROLEUM OIL	OTH	80	35192	21428
77	FLUCARBAZONE	OTH	78	45	27
78	SAFLUFENACIL	OTH	72	36	50
79	CLOTHIANIDIN	OTH	71	71	68
80	AMINOPYRALID	OTH	67	26	19

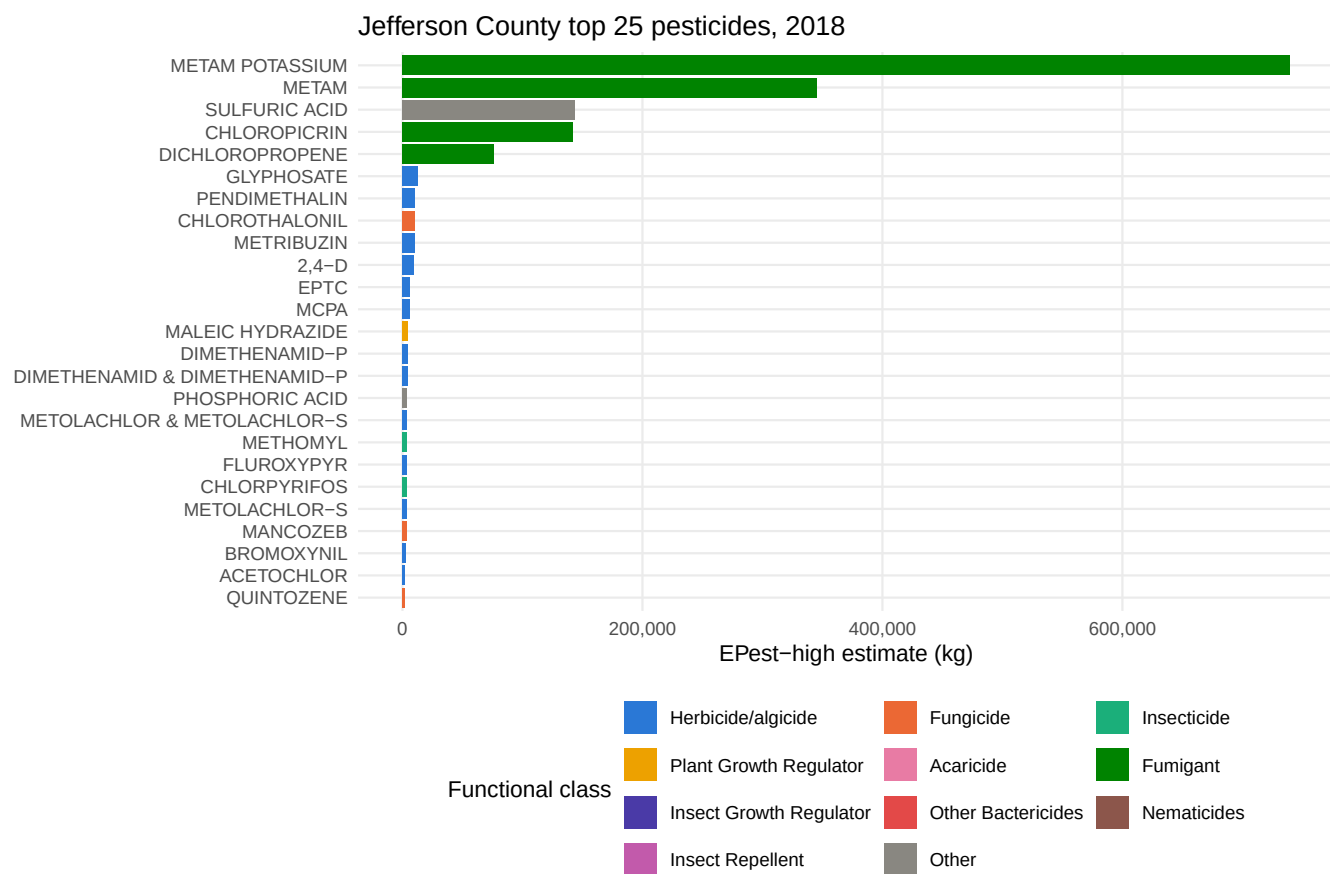


Figure 1: Top 25 compounds in Jefferson County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Jefferson County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM POTASSIUM	739900	739900
2	METAM	345594	345594
3	SULFURIC ACID	143804	143804
4	CHLOROPICRIN	1249	142419
5	DICHLOROPROPENE	76178	76178
6	GLYPHOSATE	11931	12681
7	PENDIMETHALIN	8139	10572
8	CHLOROTHALONIL	10462	10462
9	METRIBUZIN	10096	10098
10	2,4-D	9090	9177
11	EPTC	6401	6401
12	MCPA	6039	6039
13	MALEIC HYDRAZIDE	0	4978
14	DIMETHENAMID & DIMETHENAMID-P	3959	4183
15	DIMETHENAMID-P	3959	4183
16	PHOSPHORIC ACID	3914	3914
17	METOLACHLOR & METOLACHLOR-S	3316	3806
18	METHOMYL	24	3664
19	FLUROXYPYR	3608	3627
20	CHLORPYRIFOS	3475	3501
21	METOLACHLOR-S	3019	3460
22	MANCOZEB	3450	3445
23	BROMOXYNIL	2990	2990
24	ACETOCHLOR	0	2127
25	QUINTOZENE	0	1973
26	PYRIMETHANIL	1838	1838
27	DIQUAT	1638	1638
28	ATRAZINE	8	1547
29	PROPARGITE	0	1389
30	GLUFOSINATE	1339	1339
31	BOSCALID	1268	1268
32	PINOXADEN	1242	1242
33	HEXAZINONE	0	1222
34	AZOXYSTROBIN	1154	1154
35	IMIDACLOPRID	1090	1090
36	DIURON	0	1022
37	PARAQUAT	2	995
38	DICAMBA	154	917
39	FLUOPYRAM	874	874
40	DIMETHOATE	755	839
41	PYRACLOSTROBIN	782	794
42	OXAMYL	716	716
43	TRIFLURALIN	0	550
44	FLUTOLANIL	444	444
45	PROPICONAZOLE	403	403
46	CLOPYRALID	353	374
47	METOLACHLOR	297	346
48	MEFENOXAM	333	333
49	CYFLUTHRIN	0	320
50	CARFENTRAZONE-ETHYL	0	319

Table 4: Top 50 Jefferson County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	512798	512798
2	METAM POTASSIUM	339268	339268
3	SULFURIC ACID	270539	270539
4	DICHLOROPROPENE	120436	120436
5	CHLOROPICRIN	7935	63096
6	PETROLEUM OIL	770	35192
7	ETHOPROPHOS	0	23466
8	PROPAMOCARB HCL	0	20425
9	GLYPHOSATE	11686	11942
10	PENDIMETHALIN	7556	10943
11	METRIBUZIN	9762	9763
12	CHLOROTHALONIL	9687	9687
13	EPTC	8101	8101
14	2,4-D	5826	6378
15	DIMETHENAMID & DIMETHENAMID-P	4235	4934
16	MCPA	4418	4425
17	DIMETHENAMID-P	4235	4401
18	PHOSPHORIC ACID	3834	3834
19	MANCOZEB	3674	3673
20	MALEIC HYDRAZIDE	0	3264
21	METOLACHLOR & METOLACHLOR-S	2553	3153
22	METHOMYL	18	3080
23	FLUROXYPYR	3048	3061
24	METOLACHLOR-S	2444	2772
25	DIMETHOATE	1239	2510
26	BROMOXYNIL	2468	2475
27	CHLORPYRIFOS	2229	2376
28	ATRAZINE	21	1977
29	NALED	0	1912
30	PARAQUAT	33	1718
31	ETHALFLURALIN	0	1620
32	DIMETHENAMID	0	1597
33	DIQUAT	1576	1576
34	PYRIMETHANIL	1570	1570
35	PROPARGITE	0	1389
36	BOSCALID	1324	1324
37	HEXAZINONE	92	1284
38	DIURON	72	1239
39	FLUTOLANIL	1129	1129
40	PINOXADEN	1111	1111
41	QUINTOZENE	110	1097
42	IMIDACLOPRID	1080	1080
43	AZOXYSTROBIN	993	993
44	ACETOCHLOR	0	901
45	TRIFLURALIN	0	828
46	SPIROMESIFEN	0	785
47	FLUOPYRAM	728	728
48	OXAMYL	686	686
49	GLUFOSINATE	683	683
50	MALATHION	211	644

Table 5: Top 50 Jefferson County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	531684	531684
2	SULFURIC ACID	295180	295180
3	METAM POTASSIUM	236124	236124
4	DICHLOROPROPENE	106961	106961
5	CHLOROPICRIN	5962	47333
6	PETROLEUM OIL	772	21428
7	PROPAMOCARB HCL	0	20425
8	ETHOPROPHOS	0	15954
9	GLYPHOSATE	13343	13496
10	PENDIMETHALIN	6998	9312
11	METRIBUZIN	8843	8845
12	CHLOROTHALONIL	8574	8574
13	EPTC	8137	8322
14	MANCOZEB	6729	6728
15	PHOSPHORIC ACID	6128	6128
16	MCPA	5920	5940
17	2,4-D	5227	5635
18	PARAQUAT	32	4376
19	DIMETHENAMID & DIMETHENAMID-P	3525	4009
20	MALEIC HYDRAZIDE	16	3739
21	DIMETHENAMID-P	3525	3690
22	BROMOXYNIL	2980	3275
23	FLUROXYPYR	2908	2916
24	METOLACHLOR & METOLACHLOR-S	2125	2901
25	DIMETHOATE	745	2688
26	METOLACHLOR-S	2059	2424
27	OXAMYL	2388	2388
28	METHOMYL	16	2381
29	PROPARGITE	2	2135
30	CHLORPYRIFOS	1986	2133
31	TERBACIL	0	1968
32	NALED	0	1912
33	DIQUAT	1892	1892
34	ETHALFLURALIN	0	1620
35	DIMETHENAMID	0	1597
36	PYRIMETHANIL	1506	1506
37	BOSCALID	1370	1370
38	IMIDACLOPRID	1228	1228
39	ATRAZINE	18	1226
40	FLUTOLANIL	1108	1108
41	QUINTOZENE	110	1097
42	DIURON	377	1077
43	PINOXADEN	1041	1046
44	AZOXYSTROBIN	1045	1045
45	ACETOCHLOR	0	1008
46	HEXAZINONE	134	850
47	SODIUM CHLORATE	730	730
48	SULFUR	284	704
49	PYRACLOSTROBIN	629	658
50	MALATHION	208	638